R6.8 Transport-oracle census: corrected status

2026-10-20 review: the proposed general simulation is not established. The previous proof and closure claim are withdrawn. The original PDF and source are preserved in `historical-04c3e817/` with SHA-256 hashes. The recorded census values and plots are unchanged.

Counterexample to Lemma 5

In an order-5 cyclic subgroup with generator P, take the identity map, $S = \{O, P\}$ and exactly two summands. The original sums are O, P and $2P$. Removing O from the image base leaves $\{P\}$ and only the sum $2P$. The decomposition $P = O + P$ disappears. Thus the claimed sumset inclusion, count monotonicity and size lower bound do not hold as written.

The executable `review_counterexample.py` checks this enumeration and a separate kernel-fiber example. Keeping the identity repairs this example, but does not establish the theorem. A corrected proof must also account for all lifts, challenge-adaptive transport paths and their complete costs.

What the census establishes

Eight raw census files cover four curves, two base sizes and two arities: 16 cells. The programmed falsifier fires in 0 of 16 cells. Union/best class-factor-per-column ratios are 0.02-0.27. These are finite measurements; they do not establish the general simulation, its overhead, or a constant-only asymptotic effect. The derived cost uses the recorded model, not end-to-end wall time. No ledger row is promoted.

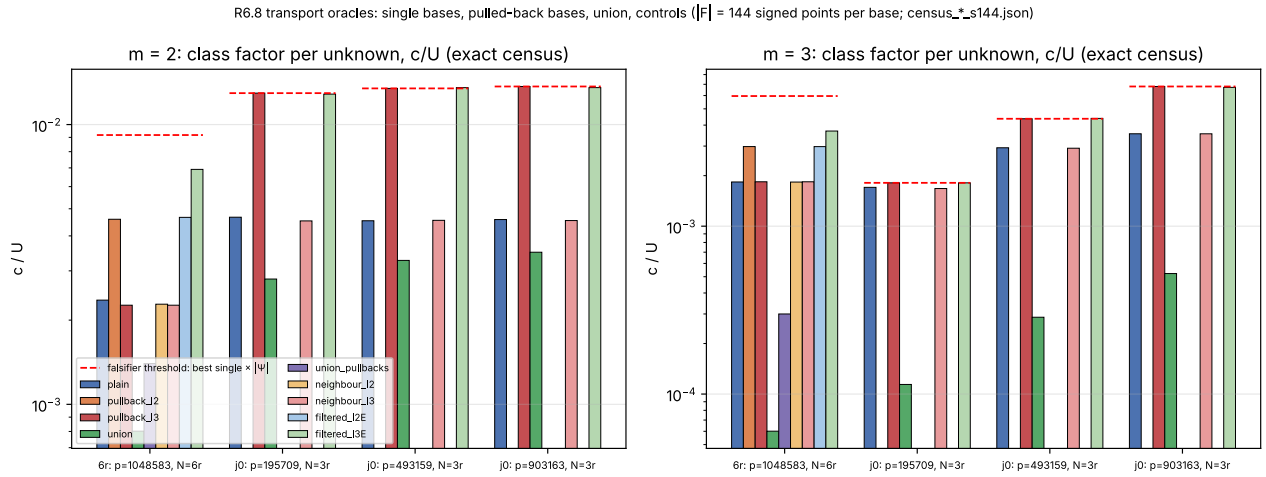


Figure 1: Unchanged recorded class factors. Finite census only; no general proof follows.

Recorded cost comparison

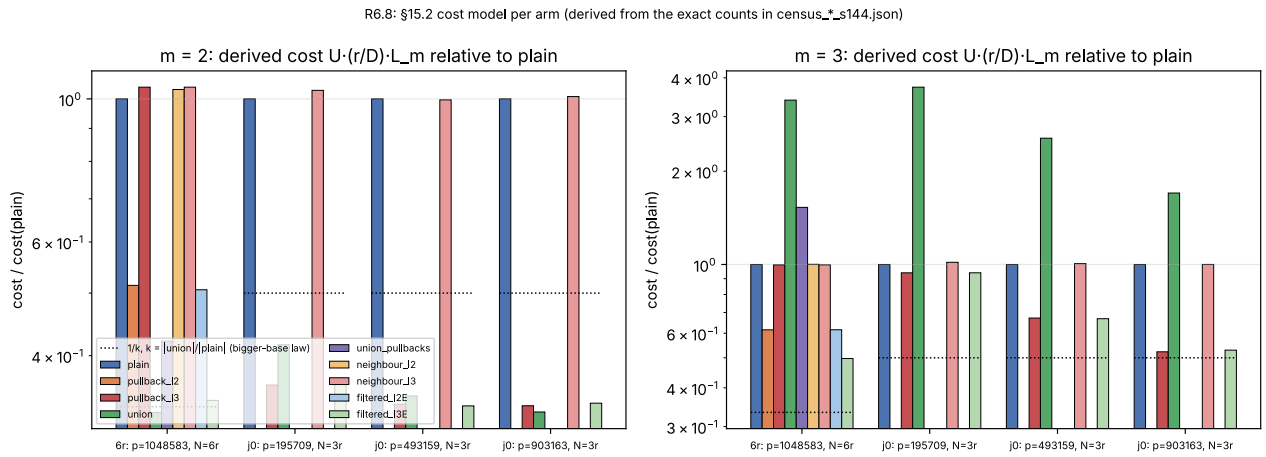


Figure 2: Unchanged derived model cost relative to the plain base.

Limits and remaining obligations

Four curves, one seed and two base sizes; some three-term sumsets approach subgroup saturation. The class-factor model is heuristic. The unrestricted 6r image counter is not a subgroup-restricted identity certificate.

A corrected theorem must define identity and kernel handling, fixed versus at-most-m decompositions, lift enumeration, adaptive path dependence, and complete cost/scaling bounds. Full details and unchanged tables are in `RESULTS.md`. The historical report is retained as superseded evidence.